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Testing Base for Quadruped Robot's Legs Based on Impedance Control

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Abstract - Agriculture in Ecuador faces significant challenges due to the country's mountainous geography, which makes access to certain areas difficult and affects the efficiency of agricultural practices. This project proposes the development of a quadruped robot capable of navigating these difficult terrains and performing tasks that are complicated for humans. The main objective is to build a robust test platform for the robot legs, focusing on impedance control to adapt to different terrain conditions. Through simulations and experimental tests, it was possible to optimize the impedance control parameters, improving the robot's ability to interact with its environment. The results indicate that the robot can operate effectively in challenging terrain, suggesting significant potential for improving efficiency and safety in agricultural operations. In conclusion, the implementation of this quadruped robot could not only reduce risks for farmers, but also contribute to more sustainable agricultural practices in Ecuador.

I. INTRODUCTION

Agriculture is one of the main economic pillars in Ecuador, characterized by a wide variety of crops grown throughout the country [1]. However, most of these crops are located on uneven mountainous terrains, which present significant challenges in terms of efficiency and maintenance. Farmers often face difficulties accessing certain areas, which hinders optimal agricultural practices. To address these issues, we propose the development of a quadruped robot capable of navigating these rugged environments and performing tasks that are challenging for humans. Our project focuses on analyzing the dynamics of a quadruped leg to create a model that can estimate the torques required by the motors to jump to a certain height with specific physical characteristics. This platform will also help in testing and selecting suitable sensors for our applications. Our innovation aims to improve efficiency and safety in agriculture, minimize human risk in hazardous terrains, and support sustainable agricultural practices.

II. PROBLEM STATEMENT

The agriculture in Ecuador is one of the biggest, with a diverse range of crops grown across the country, most of them are located on uneven mountainous terrains. The farmers face some efficiency and terrain maintenance difficulties like access to certain areas. We propose the development of a quadruped robot capable of navigating these environments and performing specific tasks that would be hard for humans to do. Our initial goal is to build a robust test platform for the robot legs, to study and design an impedance control to ensure that it can cope with the environmental challenges present in our landscapes and to study how we can configure different stiffness and damping values to suit different terrains. It can also help us test and select an adequate sensor for our application. This innovation aims to improve the efficiency and safety of farming operations, lower human risky interactions and support sustainable agricultural practices.

III. LITERATURE REVIEW

Extensive research has been conducted to address these challenges, with notable contributions in areas of robot design, control algorithms and simulation tools. For instance, [2] presented a new torque-controlled quadruped robot system based on a low-cost actuator module, ideal for impedance and force control. They carried out a systematic characterization of impedance in the foot in both static and dynamic contexts, achieving a maximum leg stiffness without active damping comparable to human leg stiffness during running. In addition, they developed an innovative controller that integrated contact forces calculated by a kino-dynamic optimizer with impedance control of the center of mass and base orientation. Torque measurement at the motor revealed differences in the resulting torque at the end effector due to gearing losses, structural flexibility of the belt and leg, and inertial losses. Despite these limitations, the experimental results confirmed the effectiveness of impedance control. In experiments with a 2-degree-of-freedom leg, a stiffness of 266 N/m was measured, corresponding to a dimensionless stiffness of 10.8, comparable to the stiffness of the human leg.

There is also research in optimization methods to change the impedance control parameters [3]. In this work presents a locomotion controller for quadruped robots that integrates Cartesian impedance control with Quadratic Programming (QP) to optimize robot performance and compliance in challenging terrains. Impedance control is based on a mass-spring-damper model, allowing the robot to respond to external disturbances and model errors, which is crucial for maintaining stability. The QP formulation is used to calculate optimal torques that align with the desired values of the impedance control, while satisfying physical constraints such as torque limits and friction conditions. This approach enables a coupling strategy that relaxes trajectory tracking when constraints are close to being violated, which is essential in extreme environments. Compared to hierarchical quadratic control (HQP), which requires the solution of multiple optimization problems and can be computationally intensive, the proposed method simplifies the process by solving only one QP, thus improving computational efficiency. Experiments were conducted with the ANYmal robot, which has 12 joints actuated by a series elastic actuator and a 400 Hz control cycle, demonstrating that the proposed controller is not only more efficient, but also offers better performance in external disturbance estimation and model error compensation, highlighting its effectiveness in adaptive locomotion under difficult conditions

There are also works related to the challenge of controlling the leg force and the position leading to impedance control as a solution [5]. In this paper the authors highlight impedance control as an effective strategy to improve stability and adaptability during jogging on uneven terrain. This approach involves dynamic modulation of the mass (M), spring (K), and damper (B) parameters of the robot legs,

adjusting them as a function of the five phases of motion: lifting, landing, unsynchronized, weight acceptance, and pushing. The estimation and modulation of these parameters is based on a second-order system model, where the robot response is adjusted to minimize the impact of contact forces and terrain irregularities. For example, during the landing phase, the values of K and B can be reduced to allow for greater compression and shock absorption, while in the thrust phase, they can be increased to maximize propulsion. The advantages of this method include increased adaptability and stability, as it allows the robot legs to respond flexibly to variations in the environment, which reduces the risk of falling and improves energy efficiency. However, limitations are also identified, such as the complexity in implementing the control, which requires an accurate sensor system to measure forces and positions in real time, as well as the need for proper modeling of terrain characteristics, which can be challenging in highly variable or unknown environments. This review suggests that, despite its limitations, variable impedance control represents a promising direction for the development of more robust and efficient quadrupedal robots.

Reviewing the literature on impedance control in quadrupeds, different strengths and weaknesses are observed. The first approach focuses on the innovation of torque-based impedance control, using a low-cost actuator module that allows accurate characterization of leg stiffness, similar to that of a human leg. However, this approach faces significant limitations, such as gear losses and structural flexibility, which can affect the accuracy of impedance control and thus its effectiveness in real applications. On the other hand, the second approach offers a more computationally efficient solution by integrating impedance control with quadratic programming (QP), optimizing torques to improve stability in difficult terrain. Although it simplifies the optimization process, its implementation is complex due to the need for accurate modeling of the system and its environment. Finally, the third approach is the most adaptive, allowing real-time dynamic modulation of mass, spring and damper parameters to improve stability and energy efficiency in rough terrain. However, this adaptability requires a very precise sensor system and detailed modeling, which complicates its implementation in highly variable or unknown environments.

Summarizing, an impedance control for a quadruped leg will be implemented that combines the strengths of the three approaches studied, with the objective of dynamically adjusting the stiffness and damping as a function of the joint contact force, specifically for flat surfaces. Using the accurate torque-based stiffness characterization of the first approach, different stiffness and damping settings will be set to optimize the interaction with the flat surface, ensuring accurate and stable control. The computational efficiency of the second approach will be integrated using quadratic programming (QP) to adjust the torques in real time, maintaining system stability with varying configurations. In addition, the adaptivity of the third approach will be employed for dynamic modulation of the parameters, adjusting the leg response according to surface conditions, allowing for improved stability and energy efficiency in movements over flat terrain. This combination will provide optimized impedance control, capable of adapting to different stiffness and damping configurations, maximizing stability and performance on flat surfaces.

IV. METHODOLOGY

To achieve the objectives, we aim to design an impedance control system for our quadruped leg which will be simulated with Matlab and Simulink's Simscape Multibody to test the response it could have. We are going to start by determining the kinematic model of our leg like [2]. Even though it is like the previous work we have some differences in our test base having the following configuration.

A. Kinematics

For our Kinematic model we use inverse kinematics but for the purpose of the study we also did a Denavit Hartenberg parameters table.

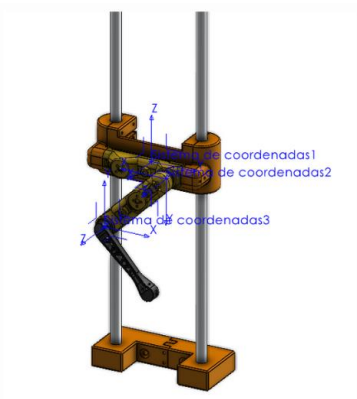


Fig. 1 Test base Transforms

DH parameters shown in the table below help us to compute the Jacobian expression. Setting an initial configuration will help us computing our initial Jacobian and check if our results are correct using Robotics Toolbox in Matlab.

TABLE I: DH Parameters

i	*a*	*α*	*d*	*θ*
0	-90°	0.0635	d1	90
1	0°	0.187	0.096	θ1
2	0°	0.198	0.058	θ2

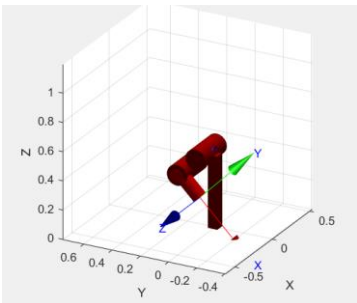


Fig. 2 DH Matlab's Robotics Toolbox

We also compute a Jacobian given an initial configuration d1=0.5, θ1=pi/4 and θ2=pi/2 with RTB's library that gave us a reasonable data.

```
-0.0000   -0.0000   -0.0000
 0.0000   -0.2722   -0.1400
 1.0000    0.0078    0.1400
      0   -1.0000   -1.0000
      0    0.0000    0.0000
      0    0.0000    0.0000
```

Fig. 3 Jacobian Matlab's Robotics Toolbox

B. Dynamics

We will apply Lagrange method [6] to find the dynamic equation by modeling our leg as a mass-spring-damper system that will help us to predict the dynamic behavior of our leg subjected to external forces as shown in the fig 2:

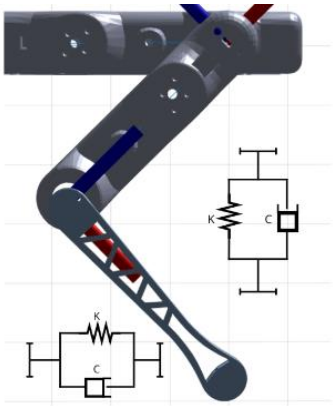


Fig. 4 Dynamic Model

To determine the model, we will work at the level of energies of the system, considering the kinetic energy, powers and those caused by non-conservative forces.

Kinetic Energy (K):

$$K = \frac{1}{2}m_1 v_1^2 + \frac{1}{2}I_1\dot{\theta}_1^2 + \frac{1}{2}m_2 v_2^2 + \frac{1}{2}I_2\left(\dot{\theta}_2 + \dot{\theta}_1\right)^2 \ (1)$$

Potencial Energy (U):

$$U = m_1 g lc_1(1 - \cos(\theta_1)) + m_2 g(l_1 + lc_2 - (l_1\cos(\theta_1) + l_2\cos(\theta_1 + \theta_2))) \ (2)$$

Lagrangian (L):

$$L = K - U \ (3)$$

Euler-Lagrange Equations:

$$\tau_1 = \frac{d}{dt}\left(\frac{dL}{d\dot{\theta}_1}\right) - \frac{dL}{d\theta_1} \ (4)$$

$$\begin{aligned}\tau_1 = &-(l_1 lc_2 m_2 \sin(\theta_2))\dot{\theta}_2{}^2 - \left(2l_1 lc_2 m_2 \sin(\theta_2)\dot{\theta}_2\right) \\ &+ \ddot{\theta}_1\left(m_2 l_1^2 + 2m_2\cos(\theta_2)\, l_1 lc_2 + m_1 lc_1{}^2 + m_2 lc_2{}^2 + I_1 + I_2\right) \\ &+ \ddot{\theta}_2(m_2 lc_1^2 + I_1 m_2\cos(\theta_2)\, lc_2 + I_2) \\ &+ gl_1 m_2 \sin(\theta_1 + \theta_2) \\ &+ glc_1 m_1 \sin(\theta_1) \ (5)\end{aligned}$$

$$\tau_2 = \frac{d}{dt}\left(\frac{dL}{d\dot{\theta}_2}\right) - \frac{dL}{d\theta_2} \ (6)$$

$$\begin{aligned}\tau_2 = &\ \ddot{\theta}_2(m_2 lc_2^2 + I_2) + \ddot{\theta}_1(m_2 lc_1^2 + I_1 m_2\cos(\theta_2) lc_2 + I_2) \\ &+ glc_2 m_2 \sin(\theta_1 + \theta_2) \\ &+ I_1 lc_2 m_2 \dot{\theta}_1 \sin(\theta_2) \ (7)\end{aligned}$$

Obtaining a system of equations with the following form:

$$\tau = M(\theta)\ddot{\theta} + C\left(\theta, \dot{\theta}\right)\dot{\theta} + u(\theta)$$

Where:

$$M = \begin{pmatrix} m_2 l_1^2 + 2m_2\cos(\theta_2)\, l_1 lc_2 + m_1 lc_1{}^2 + m_2 lc_2{}^2 + I_1 + I_2 & m_2 lc_1^2 + I_1 m_2\cos(\theta_2)\, lc_2 + I_2 \\ m_2 lc_1^2 + I_1 m_2 \cos(\theta_2) lc_2 + I_2 & m_2 lc_2^2 + I_2 \end{pmatrix}$$

$$C = \begin{pmatrix} 0 & -(l_1 lc_2 m_2 \sin(\theta_2))\theta_2 - (2l_1 lc_2 m_2 \sin(\theta_2)) \\ I_1 lc_2 m_2 \dot{\theta}_1 \sin(\theta_2) & 0 \end{pmatrix}$$

$$u = \begin{pmatrix} gl_1 m_2 \sin(\theta_1 + \theta_2) + glc_1 m_1 \sin(\theta_1) \\ glc_2 m_2 \sin(\theta_1 + \theta_2) \end{pmatrix}$$

C. Cartesian Impedance Control

We are going to use a cartesian impedance control model to regulate the stiffness and damping of the foot related to the base of the hip, keeping the dynamics (Fig 2). The control equation is the following:

$$\tau = J^T\left(K(xd - x) - D\left(\dot{xd} - \dot{x}\right)\right) \ (5)$$

Where J is the Jacobian, K is the stiffness and D is the damping of the dynamic model. To determine the values of k, the system was simulated with values of K = 0 and D = 0, to analyze the behavior and determine the natural frequency ω_n. Once the natural frequency is obtained, we obtain K from the following equation:

$$K_0 = m * \omega_n{}^2 \ (6)$$

To determine the damping value (D), it was estimated from the logarithmic decay, which is a measure of how fast the system's oscillations fall. This decrement was calculated graphically from the natural response of the system, identifying the relationship between the amplitudes of successive peaks. Which in turn is also related to the spring compression than the impedance values of the system:

$$K_0 = \frac{mg}{\delta z} \ (7)$$

The key in this equation is the δz which refers to how much the leg compresses or deforms in the vertical direction (z-coordinate) when it's in contact with the ground and the robot is at rest (steady-state). We also know from [5] that the damping of a system can be determined by the following equation:

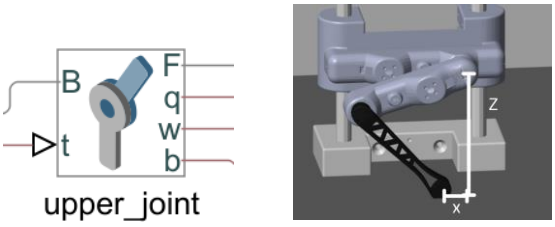
$$D = 2\zeta\sqrt{m * K} \ (8)$$

Keeping in mind that m is the mass of our system, and we only have one damping matrix for a single leg. With our values of damping and stiffness we can test now our control using Matlab and Simulink's simscape multibody library.

Before testing it is important to mention that we exported a model designed previously in SolidWorks as an urdf file that will be later imported to Simscape Multibody. It is important to stablish a correct model in the urdf to avoid simulation issues in Simscape.

For the impedance control implementation, we apply the robotics system toolbox which gives us the tools to formulate our desired equation. We used the following blocks

- Jacobian
- Mass
- Velocity Product
- Gravity



(a)

Fig. 5 (A) Simscape Multibody Joint Block (B) Cartesian coordinates

To build our equations we also use the sensing function that simscape gives us in the joint's blocks (Fig 5(A).) we need the following data:

- q1, q2: Joint's positions (q)
- qd1, qd2: Joint's velocities (w)
- qdd1, qdd2: Joint's accelerations (b)
- x = cartesian position in X axis wrt to EE
- z = cartesian position in Z axis wrt to EE
- vx= cartesian velocity in X axis wrt to EE
- vz = cartesian velocity in Z axis wrt to EE

For the cartesian coordinates measurements we can use inverse kinematics or use Simscape tools that help you measure the desired distance. In our case we choose the second option with a Transform Sensor block, this will allow us to measure between two frames of interest. I chose to measure between the EE and the hip (Fig 5(B).) in the cartesian space as mentioned above.

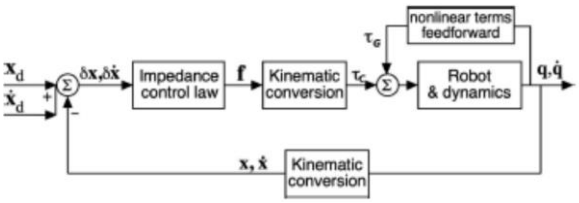


Fig. 6 Impedance Control Block Diagram

Now we have all the tools to build our [7][8] impedance law control block (Fig 6) on Matlab. It is necessary to point out that we needed an extra library for the contact between the floor surface and the EE, in this case the tip of our leg. We use Simscape Multibody Contact-Forces Library.

We will measure the response to different stiffness and damping values to check the best behavior on the values obtained so for a later work we can stablish an adaptive impedance to different situations keeping the best behavior on every situation

V. RESULTS

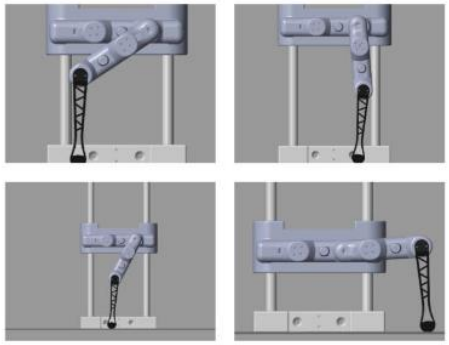


Fig. 7 Drop simulation of the leg

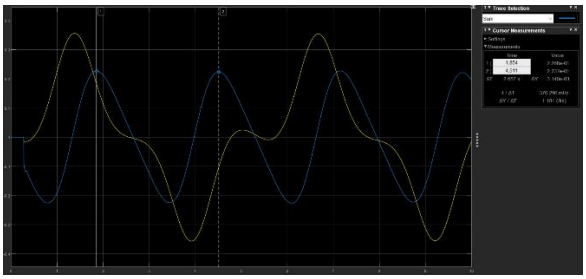


Fig. 8 Velocity response of the system in the x- and z-direction with values K= 0 and D= 0

We did a couple of iterations for some K and C values in our dynamic model that stabilizes the system. The results of this iterations were plotted in matlab and show some interesting behaviors analized in the discussion section.

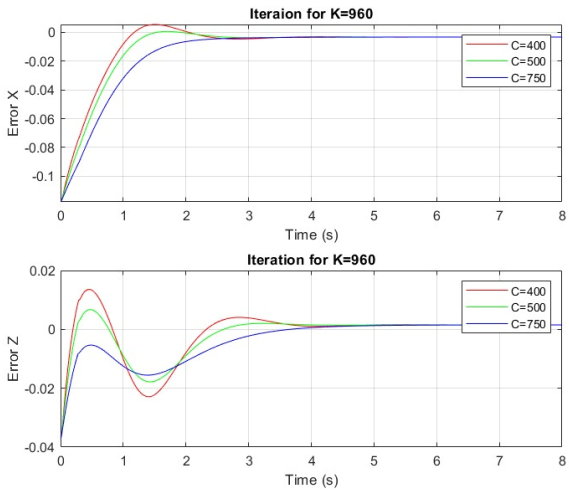


Fig. 9 Position errors in X and Z for the first Iteration

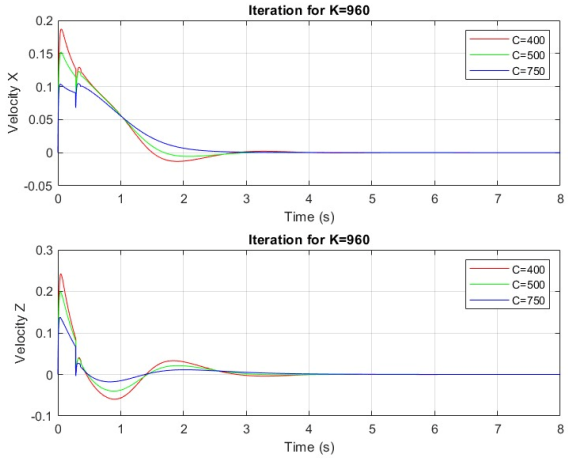


Fig. 10 Velocity errors in X and Z for the first Iteration

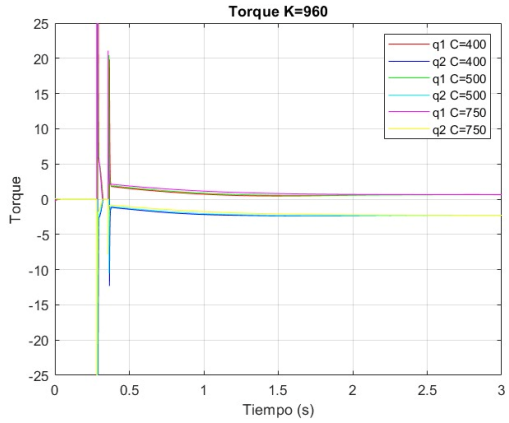


Fig. 11 Torques for q1 and q2 for the first Iteration

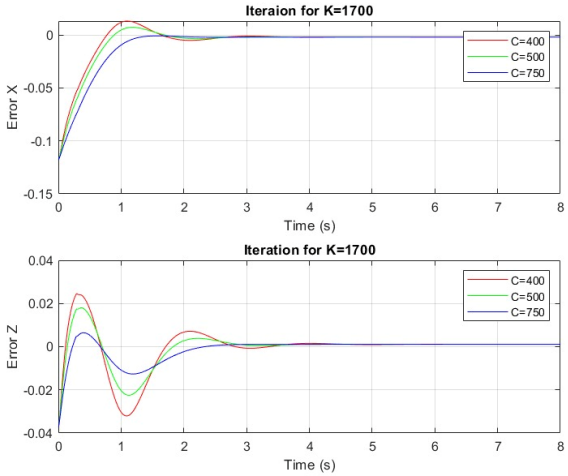


Fig. 12 Position errors in X and Z for the Second Iteration

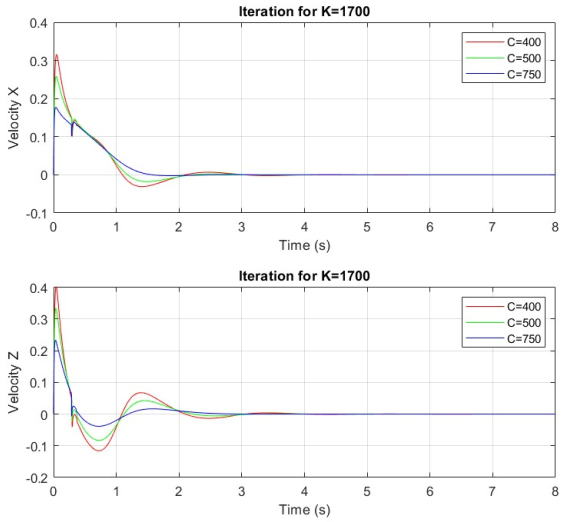


Fig. 13 Velocity errors in X and Z for the first Iteration

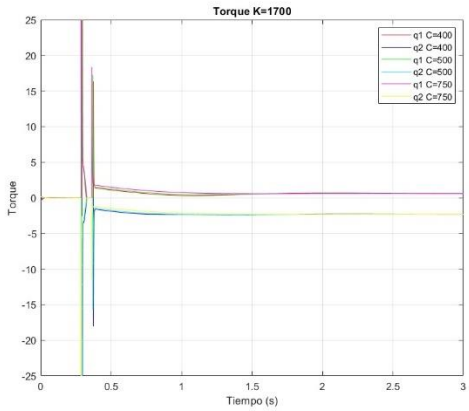


Fig. 14 Torques for q1 and q2 for the Second Iteration

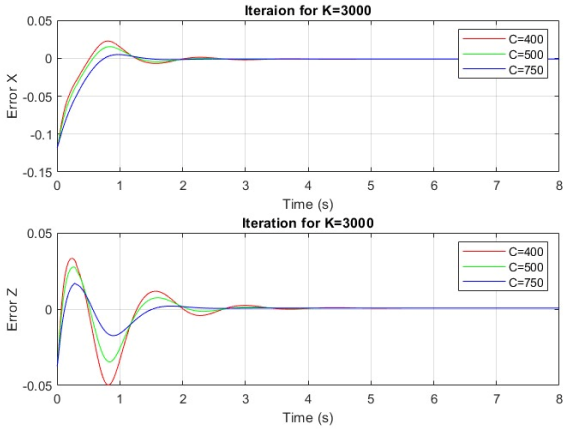


Fig. 15 Position errors in X and Z for the Third Iteration

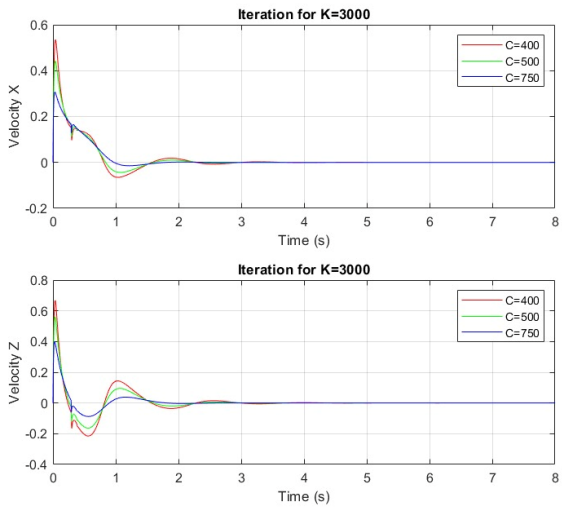


Fig. 16 Velocity errors in X and Z for the Third Iteration

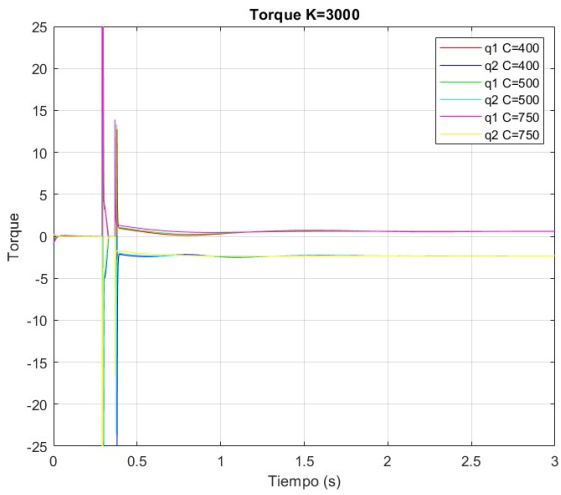


Fig. 17 Torques for q1 and q2 for the Second Iteration

VI. DISCUSSIONS

Analyzing the figures in results section, we can see that the system response has an oscillatory behavior, with period T=2.658 s, which results in a natural frequency of $\omega_n = 2.363$. Estimating the values of K and D from the representative values of the system behavior, we obtained the following values.

Analyzing the different graphs in figures a and b, we can notice that, as K increases, the system becomes stiffer, meaning it resists deviations from the desired position more strongly. This results in a faster response to positional commands, as the torque generated by the controller in response to deviations is higher.

The system's ability to correct for errors improves, allowing the joint to reach the desired position more quickly. However, this increased stiffness comes at a cost. Higher values of K tend to induce oscillations around the desired position. This happens because the system, while correcting for deviations, may overshoot due to the high torque response, leading to a cyclical behavior where the joint continuously oscillates around the target position. These oscillations can lead to higher peak torques in the joints, which may cause mechanical stress and reduce the overall efficiency of the control.

This increase in torque not only demands more power from the actuators, but also leads to higher energy consumption, which can reduce the overall efficiency of the system. In addition, continuous operation at high torque levels can lead to overheating of the motors and premature wear of mechanical components, compromising the durability and reliability of the system. Therefore, it is crucial to find a balance between stiffness and damping to minimize these adverse effects, optimizing energy efficiency without sacrificing control accuracy and stability.

To counteract the oscillations induced by high stiffness values, damping D must be sufficiently high. Damping works by applying a torque proportional to the velocity of the joint, which opposes the motion. This opposing torque reduces the amplitude of the oscillations, leading to a smoother and more stable approach to the desired position. However, too much damping can slow down the response of the system, making it sluggish.

There is a delicate balance between K and D that needs to be achieved. If K is too high without sufficient damping, the system will oscillate excessively, leading to high torques that could potentially damage the joint actuators or lead to instability. Conversely, if D is too high, the system's response may be overdamped, leading to a slower convergence to the desired position, which may be undesirable in applications requiring quick reactions.

For the first iteration case (Fig 9) we have K=960 and C=400, 500 and 750 considering the curves for each we can conclude that the behavior of the control reduce the oscillations with a higher C. The same case is for the velocity errors (Fig 10). Finally, we can check a decrease in torque for a joint and an increase in torque for the other joint (Fig 11) (Fig 14). This behavior is repeated between iterations creating a pattern of how it will respond to external forces. We can use this data to later create an algorithm of adaptative control.

VII. CONCLUSIONS & RECOMMENDATIONS

CONCLUSIONS

It was possible to design an impedance control system that properly balances the values of stiffness (K) and damping (D), allowing the quadruped robot to quickly reach the desired positions without compromising the stability of the system. Although a high value of K increases the speed of response, the introduction of adequate damping was crucial to mitigate oscillations and ensure stable and efficient joint behavior. We also meet the desired position and torque targets without the need to monitor its values in real time simplifying the implementation and with an optimal performance.

Careful adjustment of the values of K and D made it possible to reduce torque peaks that could cause mechanical damage or excessive energy consumption. This approach contributes to increased actuator durability and system energy efficiency, which is especially important in mobile applications where energy resources are limited.

RECOMMENDATIONS

- The torques generated by the joints must be carefully controlled, especially when dealing with high stiffness values. High torques can lead to overheating of the actuators, increased wear and potentially damage the mechanical structure of the robot.
- Avoid very high torques since they imply higher energy consumption, as more power is required to generate the necessary forces to achieve the desired motion.
- Avoid setting positions that the leg won't be able to reach to avoid simulation errors. Add a validation to avoid this by inverse kinematics.

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